long-lost sibling. Second, you are not alone in having this reaction to quantum mechanics. It is a view held to a greater or lesser extent by some of the most revered physicists of all time. Einstein refused to accept quantum mechanics fully. And even Niels Bohr, one of the central pioneers of quantum theory and one of its strongest proponents, once remarked that if you do not get dizzy sometimes when you think about quantum mechanics, then you have not really understood it.

It’s Too Hot in the Kitchen

The road to quantum mechanics began with a puzzling problem. Imagine that your oven at home is perfectly insulated, that you set it to some temperature, say 400 degrees Fahrenheit, and you give it enough time to heat up. Even if you had sucked all the air from the oven before turning it on, by heating its walls you generate waves of radiation in its interior. This is the same kind of radiation—heat and light in the form of electromagnetic waves—that is emitted by the surface of the sun, or a glowing-hot iron poker.

Here’s the problem. Electromagnetic waves carry energy—life on earth, for example, relies crucially on solar energy transmitted from the sun to the earth by electromagnetic waves. At the beginning of the twentieth century, physicists calculated the total energy carried by all of the electromagnetic radiation inside an oven at a chosen temperature. Using well-established calculational procedures they came up with a ridiculous answer: For any chosen temperature, the total energy in the oven is infinite.

It was clear to everyone that this was nonsense—a hot oven can embody significant energy but surely not an infinite amount. To understand the resolution proposed by Planck it is worth understanding the problem in a bit more detail. It turns out that when Maxwell’s electromagnetic theory is applied to the radiation in an oven it shows that the waves generated by the hot walls must have a whole number of peaks and troughs that fit perfectly between opposite surfaces. Some examples are shown in Figure 4.1. Physicists use three terms to describe these waves: wavelength, frequency, and amplitude. The wavelength is the distance between successive peaks or successive troughs of the waves, as illustrated in Figure 4.2. More peaks and troughs mean a shorter wavelength, as they must all be crammed in between the fixed walls of the oven. The frequency refers to the